SightLine: Reference-Anchored Scene-Change Gating for Energy-Aware Assistive Vision-Language Models

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Abstract—Wearable assistive vision-language models (VLMs) can narrate the environment for blind and low-vision (BLV) users, but running a VLM on every frame drains the battery and floods the auditory channel. We present SightLine, a lightweight CPU-side gate that decides, frame by frame, whether a VLM forward pass is warranted. SightLine compares each frame against the last processed reference frame, so gradual drift accumulates, and fuses a dead-zone difference hash ($\varepsilon = 7.0$), a color histogram, a 4×4 region-of-interest (ROI) motion proxy, and a latched hazard state with cooldown and a two-rate watchdog heartbeat. On a single unannotated 20-second egocentric clip of technician benchwork, exact perceptual hashing fires on 100% of frames because of sensor noise, whereas the Endurance profile suppresses 94.7% of frames; using Jetson Orin Nano power parameters and RTX 2050 service times, this corresponds to a modeled 7.52 W and 4.33 h runtime versus 2.83 h for continuous inference. The same profile is blind to localized hazards (0% recall on a synthetic vehicle-entry scenario). The Safety-Priority profile (SightLine++: 2D dHash plus ROI max-pooling) recovers 100% synthetic hazard recall at 88.4% suppression, but on the same real clip it suppresses only 80.5% of frames and saturates the modeled GPU queue (100% duty, 2.83 h). We therefore characterize edge gating as a safety–efficiency trade-off that requires task-conditioned profiles or additional mobility sensing. Evidence is limited to synthetic scenarios, one real clip, and modeled power; we describe a planned BLV user study but report no human-subject results.

Keywords—Vision-language models, assistive technology, scene-change gating, perceptual hashing, edge computing, wearable devices, energy efficiency.

I. INTRODUCTION

Wearable assistive vision systems give BLV users situational awareness by narrating obstacles, text, and scene changes. Compact VLMs such as Florence-2 [1] make open-vocabulary description feasible on edge hardware, but streaming 30 fps video through a VLM is impractical on a wearable: it depletes the battery, produces thermal load, and generates redundant speech that can mask environmental sounds used for orientation [2].

Front-end frame filtering is well studied in server-side video analytics (NoScope [3], Focus [4], Reducto [5], Glimpse [6]). Wearable assistive vision differs in three ways that motivate this work:
1. Sensor noise on flat backgrounds: CMOS thermal and quantization noise flips individual difference-hash bits on untextured surfaces, so exact hashing never suppresses frames.
2. Gradual drift: Differencing consecutive frames ($I_t - I_{t-1}$) misses slow camera motion or slowly approaching obstacles whose per-frame change stays below threshold.
3. Auditory channel contention: Ambient narration should not delay urgent warnings; this motivates a priority-based speech queue downstream of the gate.

A central finding is that gating sensitivity and energy savings trade off against each other. A coarse global gate (Endurance profile) suppresses most frames on a quiet workbench but cannot see small localized hazards. Adding ROI max-pooling and 2D hashing (SightLine++) catches localized incursions, but on a real clip with continuous hand and tool motion it triggers often enough to saturate a single edge GPU worker. Gating is therefore a Pareto frontier, not a single setting.

Contributions:
1) A reference-anchored multi-cue gate (dead-zone dHash, color histogram, ROI motion, latched hazard state with cooldown, two-rate watchdog) and its decision policy.
2) A front-end latency bound (Lemma 1) that holds under an idle worker, and an explicit statement of why it fails under queue saturation.
3) An empirical study on a five-scenario synthetic suite (10 seeds) showing that dead-zone hashing removes noise-driven false triggers, that global gating misses localized hazards, and that low-threshold gating on flat documents is statistically indistinguishable from random sampling.
4) An exploratory real-video evaluation (one 20 s clip) with Florence-2 FP16 timing on an RTX 2050 and a clearly labeled, modeled Jetson Orin Nano energy projection.
5) An author-assigned FMEA risk ledger, an architectural comparison with WorldScribe [15], and a planned BLV evaluation protocol (not yet executed).

II. RELATED WORK

Assistive VLMs and wearables: Commercial systems such as OrCam MyEye [11], Seeing AI [12], Be My AI [21], and Envision Glasses [22] rely largely on user-triggered capture. WorldScribe [15] provides continuous, context-aware live description by combining on-device components with cloud VLMs. SightLine is complementary: it targets the on-device decision of when to invoke a VLM, and its gate could also serve as an upstream filter for cloud pipelines (not evaluated here).

Redundancy reduction in video analytics: NoScope [3] and Focus [4] use cheap filters and difference detectors before expensive models; Reducto [5] adapts filtering thresholds to analytics accuracy; Glimpse [6] combines on-device tracking with server offloading. These systems optimize server cost or bandwidth rather than wearable battery life and auditory load. Work on BLV users indicates a preference for concise, prioritized descriptions [14], and auditory attention is a limited resource [2], [10].

III. PROBLEM FORMULATION

Frames I_t $(t = 0 \ldots T-1)$ arrive at $f_{\text{cam}} = 30$ fps ($\Delta t = 33.33$ ms). A gate outputs $D_t \in \{0,1\}$; the VLM runs only when $D_t = 1$. The VLM has service time t_{vlm} and is single-worker; under busy-skip scheduling, requests that arrive while it is busy are dropped. End-to-end latency decomposes as:

$$\Delta t_{\text{E2E}} = t_{\text{gate}} + t_{\text{vlm}} + t_{\text{tts}} \qquad (1)$$

where t_{gate} is the delay from event onset to $D_t = 1$ (plus ≤ 3.5 ms compute on 160×120 thumbnails), t_{vlm} is VLM inference, and $t_{\text{tts}} = 135$ ms (30 ms earcon + 105 ms local Piper synthesis). For system modeling we use a nominal design service time $t_{\text{vlm}} = 380$ ms ($\mu \approx 2.63$ Hz). This is a design budget chosen within our measured range of 287--423 ms (Section VII-D), not a single measured value; a 515 ms nominal machine latency (380 + 135) follows from it.

IV. GATING FRAMEWORK

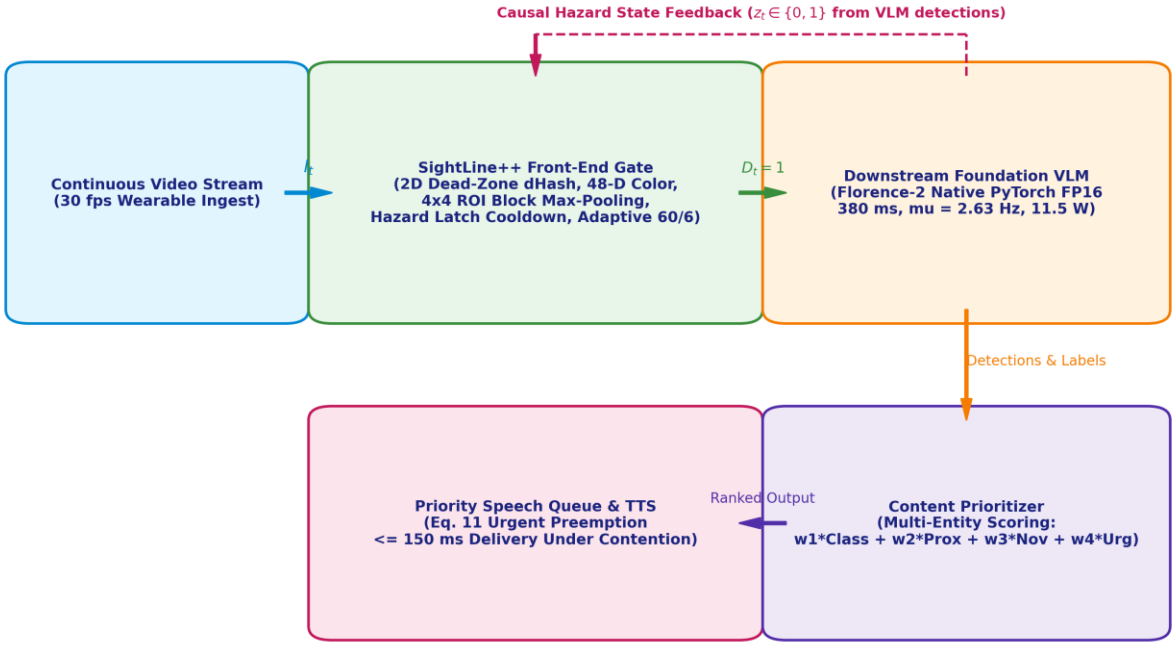


Fig. 1. System architecture. Frames reach the VLM only when the CPU gate fires; detections update the hazard state z_t that feeds back into the gate and are ranked for the speech queue.

A. Policy and Change Score

Each frame is compared against the last processed reference frame I_{t_p}, where t_p is the most recent invocation. Because the reference stays fixed across suppressed frames, gradual change accumulates until it crosses threshold τ. The change score is:

$$S_t = \alpha S_{\text{vis}} + \beta S_{\text{app}} + \gamma S_{\text{mot}} + \delta S_{\text{haz}}, \alpha + \beta + \gamma + \delta = 1 \qquad (2)$$

and the decision policy is evaluated in priority order:

$$D_t = \begin{cases} 1, & t = 0 \text{(startup)} \\ 1, & \text{else if } t - t_p \geq T_{max} \text{(watchdog)} \\ 1, & \text{else if } S_{\text{roi_max}} \geq \tau_{\text{crit}} \text{ or } S_{\text{mot}} \geq 0.20 \text{(motion override)} \\ 1, & \text{else if } S_t \geq \tau \text{ and } t - t_{\text{alert}} \geq T_{\text{cool}} \text{(nominal trigger)} \\ 0, & \text{otherwise (suppressed)} \end{cases} \quad (3)$$

Here t_{alert} is the time of the most recent invocation at which the hazard state was set ($z = 1$), and $T_{\text{cool}} = 6$ frames (200 ms).

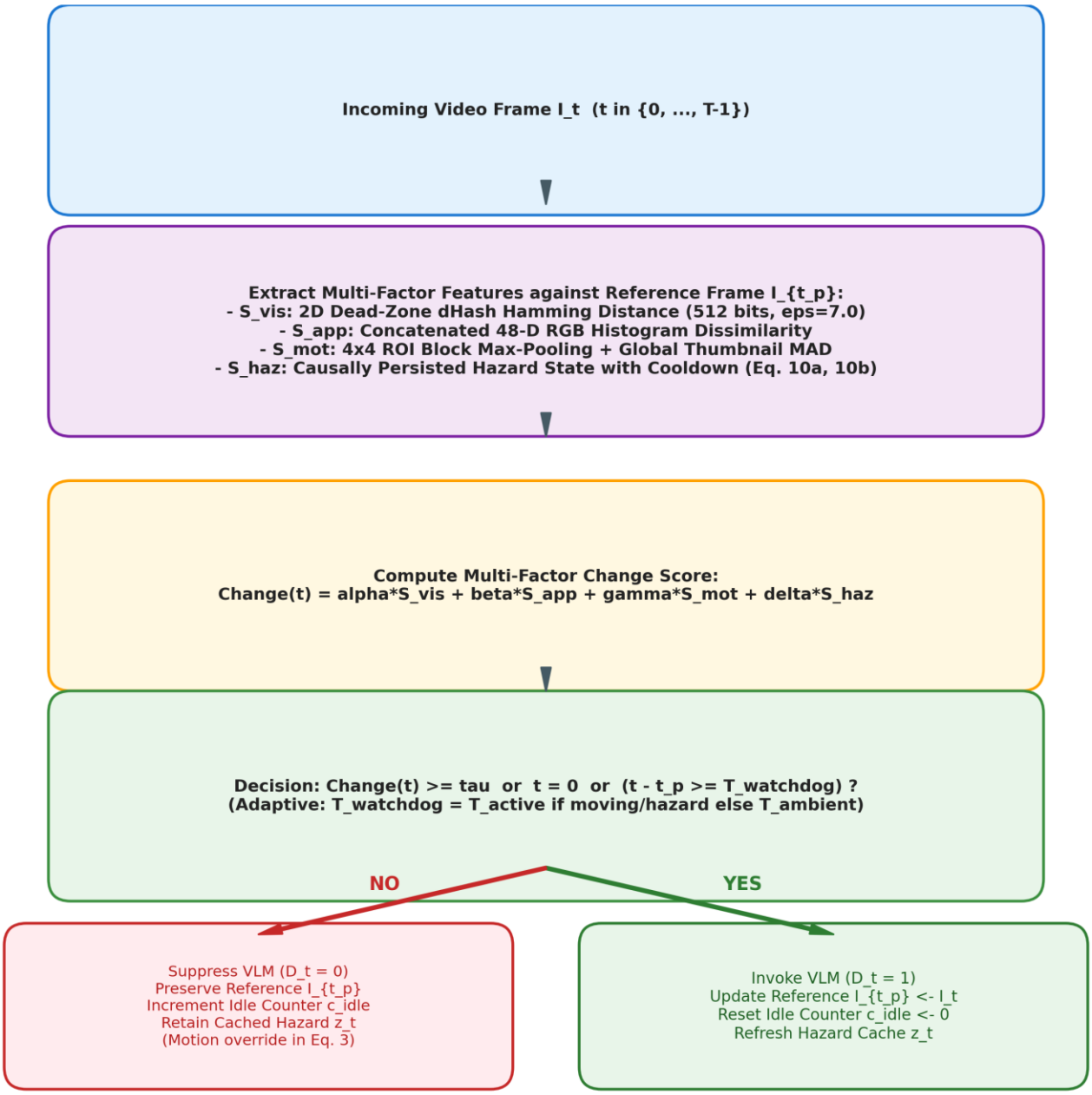


Fig. 2. Decision flow of the gate (Eq. 3). Tests run top to bottom, and the first one that matches invokes the VLM and resets the reference frame.

B. Structural Change: 1D vs. 2D Dead-Zone dHash

The frame is converted to grayscale and Lanczos-downsampled to a 17×16 grid for horizontal differences and 16×17 for vertical differences. Gradient bits use a dead zone of $\varepsilon = 7.0$ gray levels:

$$B_H(x,y) = \mathbb{1}[I(x+1,y) - I(x,y) > \varepsilon], B_V(x,y) = \mathbb{1}[I(x,y+1) - I(x,y) > \varepsilon] \quad (4)$$

On a 16×16 comparison grid per direction, the 1D hash (Endurance) uses B_H (256 bits) and the 2D hash (Safety-Priority) uses $[\, B_H \parallel B_V \,]$ (512 bits). S_{vis} is the normalized Hamming distance to the reference hash. Note that Eq. (4) sets a bit only for positive gradients exceeding ε; this one-sided definition affects the sign-invariance result discussed in Section VII-C.

C. Appearance and Motion Cues

S_{app} is the cosine distance between L2-normalized 48-D RGB histograms (16 bins per channel) of the frame and the reference. For motion, frames are reduced to 32×32 and split into a 4×4 grid of 8×8 cells; $S_{\text{roi_max}}$ is the maximum over cells of the mean absolute difference to the reference (divided by 255). $S_{\text{global_mad}}$ is the same quantity over the whole thumbnail. The motion score is $S_{\text{mot}} = S_{\text{global_mad}}$ in the Endurance profile and $0.5 \cdot S_{\text{global_mad}} + 0.5 \cdot S_{\text{roi_max}}$ in the Safety-Priority profile (SightLine++).

D. Hazard State, Cooldown, and Watchdog

After each VLM call, detections are checked against a vocabulary V_{haz} of collision hazards (car, vehicle, truck, bus, bicycle, motorcycle, scooter, stairs, curb, hole, obstacle, pole). A person counts as a hazard only if the bounding-box height fraction satisfies $h \geq 0.40$ or an image-plane centroid-velocity proxy satisfies $v \geq v_{\text{crit}}$. Under a pinhole model with a vertical field of view of $80°$ and a 1.70 m person, $h \approx 1.70/(2 \cdot 2.5 \cdot \tan 40°) \approx 0.405$ at 2.5 m, which motivates $\theta_{\text{prox}} = 0.40$. The hazard state z_t is set when any detection is a hazard, cleared when an invocation finds none, and held otherwise. The hazard score is $S_{\text{haz}} = z_t \cdot \mathbb{1}[t - t_{\text{alert}} \geq T_{\text{cool}}]$.

Caveats: (i) The velocity proxy is measured in image space, is not compensated for head motion, and is not a metric speed; the person rule is specified and implemented but untested in real crowds. (ii) In the Default, Hazard-Tracking, and Quiescent profiles $\delta \geq \tau$ (Table I), so a latched hazard alone can satisfy the nominal trigger once the cooldown expires; the cooldown therefore bounds, rather than eliminates, re-firing on a static hazard.

The watchdog guarantees a maximum inter-invocation interval: $T_{max} = T_{\text{active}}$ during motion ($S_{\text{mot}} \geq 0.02$ or $S_{\text{roi_max}} \geq 0.05$) or hazard tracking ($z_t = 1$), and T_{ambient} otherwise.

Lemma 1. Lemma 1 (Front-end gate delay): If the downstream worker is idle when a trigger arrives, the gate delay for an event at frame g is at most $(T_{max} - 1) \cdot \Delta t$.
Proof: The watchdog fires whenever $t - t_p \geq T_{max}$. If $t_p = g$ the delay is 0. Otherwise $t_p \leq g - 1$, so the next watchdog invocation is at $t_p + T_{max} \leq g + T_{max} - 1$. ■

This gives 166.7 ms for $T_{\text{active}} = 6$ and 1.97 s for $T_{\text{ambient}} = 60$. The bound does not hold under queue saturation. Because $T_{\text{active}} = 6$ corresponds to 5 Hz requests against a ≈ 2.6 Hz service rate, the bound is weakest precisely during sustained motion.

V. TASK-CONDITIONED PROFILES

Profiles set the weights, threshold, and heartbeat periods (Table I). To keep a fail-safe heartbeat in quiescent use, T_{ambient} is bounded at 150 frames (5.0 s) rather than disabled.

TABLE I: TASK-CONDITIONED PARAMETER PROFILES

Profile	alpha	beta	gamma	delta	tau	T_ambient	T_active	T_cool
Default navigation	0.25	0.20	0.30	0.25	0.18	60 (2.0 s)	6 (200 ms)	6 (200 ms)
Obstacle / hazard tracking	0.20	0.10	0.40	0.30	0.14	30 (1.0 s)	6 (200 ms)	6 (200 ms)
Paced ambient monitoring	0.30	0.30	0.20	0.20	0.22	60 (2.0 s)	12 (400 ms)	12 (400 ms)

Profile	alpha	beta	gamma	delta	tau	T_ambient	T_active	T_cool
Document inspection	0.40	0.45	0.10	0.05	0.10	150 (5.0 s)	30 (1.0 s)	15 (500 ms)
Quiescent seated rest	0.25	0.20	0.30	0.25	0.25	150 (5.0 s)	30 (1.0 s)	15 (500 ms)

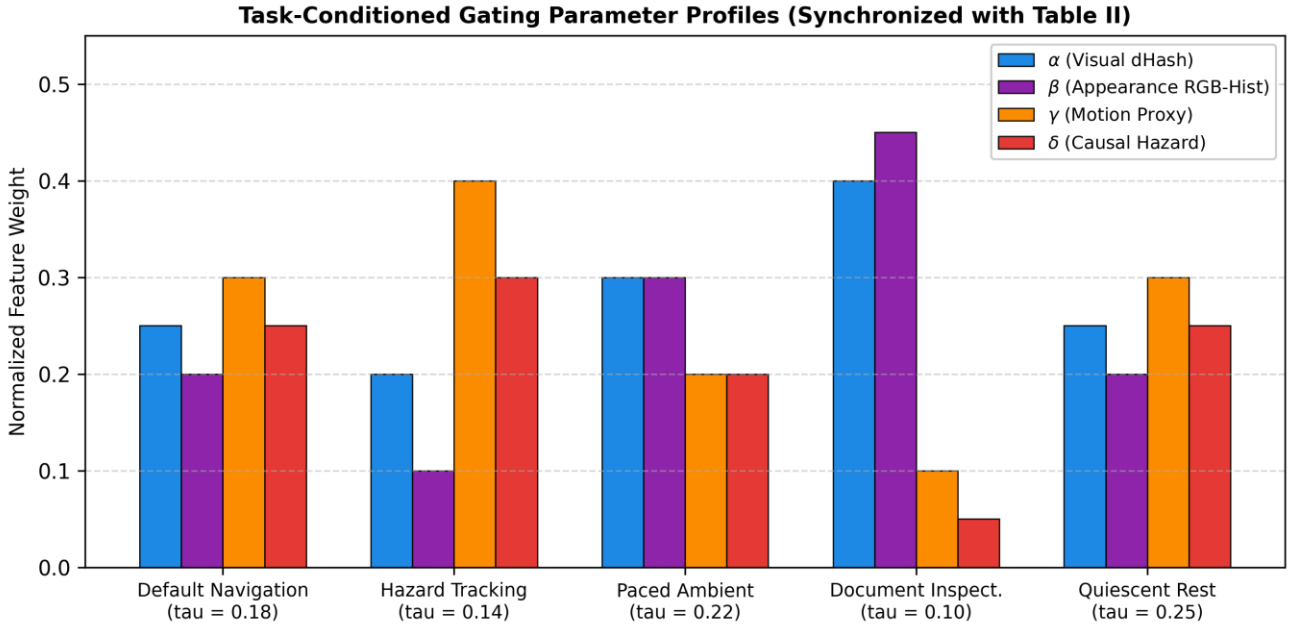


Fig. 3. Task-conditioned feature weight distributions across assistive operational modes (Default Navigation, Obstacle / Hazard Tracking, Paced Ambient Monitoring, Document Inspection, and Quiescent Seated Rest), synchronized with Table I.

VI. DOWNSTREAM SPEECH PRIORITIZATION

The gate is independent of the speech stage, but we describe the stage for completeness. Each detected entity receives:

$$\text{Priority}(o_i) = \sum_{j=1}^{4} w_j f_j(o_i), \mathbf{w} = [0.45,\ 0.30,\ 0.15,\ 0.10] \qquad (5)$$

where f_1 is hazard-vocabulary membership, $f_2 = \text{clip}(1 - d/d_{max}, 0{,}1)$ is a proximity proxy from box scale ($d_{max} = 10$ m), $f_3 = \text{clip}(v/v_{max}, 0{,}1)$ is a velocity proxy ($v_{max} = 5$ m/s), and f_4 is caption novelty. Entities with priority ≥ 0.70 preempt ongoing ambient speech (≤ 15 ms stop, 30 ms earcon, 105 ms synthesis). In an illustrative synthetic example ($f = 1.0{,}0.75{,}0.84{,}0.90$; priority 0.891), preemption delivers the warning in 135 ms versus 3.4 s behind a FIFO queue. This is a worked example, not an experimental result; the queue is not evaluated on users in this paper.

VII. EXPERIMENTS

A. Metrics and Matching

Suppression is $(1 - N_{\text{inv}}/T) \times 100\%$. A hazard event at onset g counts as detected if an invocation occurs in $[g, g + W - 1]$ with $W = 6$ frames (200 ms). W is an engineering target for the gate stage only; it is not derived from Patla [23], whose 0.5–1.0 s figures describe sighted locomotion and bound the total system budget. Pre-onset invocations and the startup frame are excluded from dynamic recall. Because synthetic scenarios are scripted and thresholds were tuned on them, 100% recall and 0 f delay should be read as consistency checks, not generalization evidence.

B. Synthetic Suite

Five scripted scenarios of 200 frames at 160×120 with Gaussian noise ($\sigma = 3.5$) are run over 10 seeds: (A) static scene with noise; (B) ambulatory drift with luminance steps; (C) a vehicle entering at frame 130; (D) a document with alternating bar contrasts (50 events); (E) gradual dimming followed by an $\times 1.8$ illumination surge.

TABLE II: SCENARIO C (VEHICLE ENTRY AT t = 130), 10 SEEDS

Configuration	Invocations	Suppression	Dynamic recall (W = 6)	Delay
dHash only (tau = 0.18)	15.7 ± 0.6	92.2%	50.0%	4.00 f
Default full gate	1.0 ± 0.0	99.5%	0.0%	n/a
Low-tau (0.05), cooldown 6	15.9 ± 0.3	92.0%	100.0%	2.30 f
Low-tau (0.05), no cooldown	68.7 ± 0.5	65.7%	100.0%	2.30 f
Watchdog T = 6, cooldown 6	34.0 ± 0.0	83.0%	100.0%	2.00 f
Watchdog T = 15, cooldown 6	20.0 ± 0.0	90.0%	100.0% fixed onset; 40.0% phase-averaged*	5.00 f
SightLine++ (2D dHash + ROI)	23.2 ± 0.4	88.4%	100.0%	0.00 f

*Under fixed onset t=130, Watchdog T=15 fires at t=135 (delay 5 f <= W=6, 100% recall), but phase-averaged recall across all 15 offsets is 40.0% (6/15).

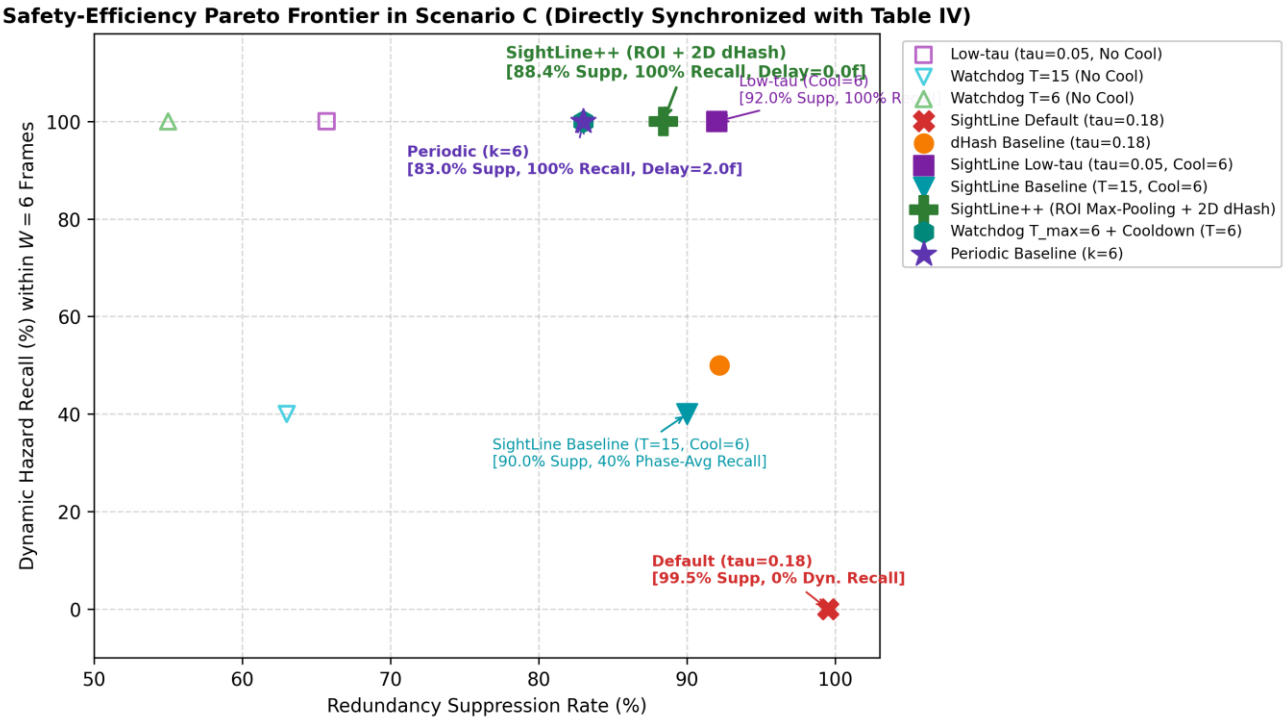


Fig. 4. Safety–efficiency trade-off on Scenario C. Points toward the upper right are better; Low-tau sits slightly above SightLine++ here but is noise-driven elsewhere.

Three observations follow: First, the Default gate suppresses 99.5% of frames but never sees the vehicle (global averaging dilutes a localized change). Second, SightLine++ achieves 100% recall at onset, but on this scenario it is not the most efficient configuration: Low-τ with cooldown reaches higher suppression (92.0%) with 100% recall. Low-τ's behavior on documents and real video is, however, noise-driven (Section VII-C), so we do not recommend it. Third, the 0.00 f delay of SightLine++ should not be attributed to ROI pooling alone: an adaptive watchdog ($T_{\text{ambient}} = 60/T_{\text{active}} = 6$) without ROI also triggers at 0.0 f here, because 10 frames had elapsed since the previous heartbeat ($130 - 120 \geq T_{\text{active}}$). Across all 60 ambient arrival phases the adaptive watchdog's delay ranges from 0 to 5 frames (mean 0.25 f, 8.3 ms). The 5-frame advantage over the fixed $T = 15$ watchdog (5.00 f → 0.00 f) is real but relative to a weak baseline. An onset-phase sweep that isolates ROI pooling from watchdog timing is needed and is listed as future work.

C. Documents, Noise, and Sign Invariance

Exact hashing ($\varepsilon = 0$) fires on every frame in noise (200 of 200 invocations in the zero-edit control); dead-zone hashing ($\varepsilon = 7.0$) reduces this to 14, a 93% suppression. With a low-threshold Text profile ($\tau = 0.10$) on Scenario D, dynamic recall is $63.9\% \pm 13.2\%$ at $85.0\% \pm 3.1\%$ suppression. However, comparing the Text profile ($61.6\% \pm 10.7\%$) with a budget-matched random sampler ($61.2\% \pm 5.2\%$) gives Welch $t = 0.12$,df $= 9.43$, $p = 0.91$, and a TOST equivalence test rejects differences of 10 percentage points or more ($p_{\text{TOST}} = 0.0098$, or $p \approx 0.015$ under conservative equal-variance assumptions). Without OCR, hash-based gating on flat documents is statistically indistinguishable from random sampling. Separately, horizontal-bar contrast swaps yield 0.0% recall once noise is dead-zoned ($\varepsilon \geq 1.5$). Because Eq. (4) is one-sided, this invariance may partly be an artifact of the thresholding; a two-sided (absolute-gradient) variant should be tested before concluding that magnitude-sensitive edge filters or OCR are required.

The threshold sweep (τ from 0.05 to 0.40, Default weights) is a step, not a smooth curve: Scenario C recall and Scenario D invocations drop sharply between $\tau \approx 0.07$ and 0.08, and recall is 0% for all larger τ including the default 0.18.

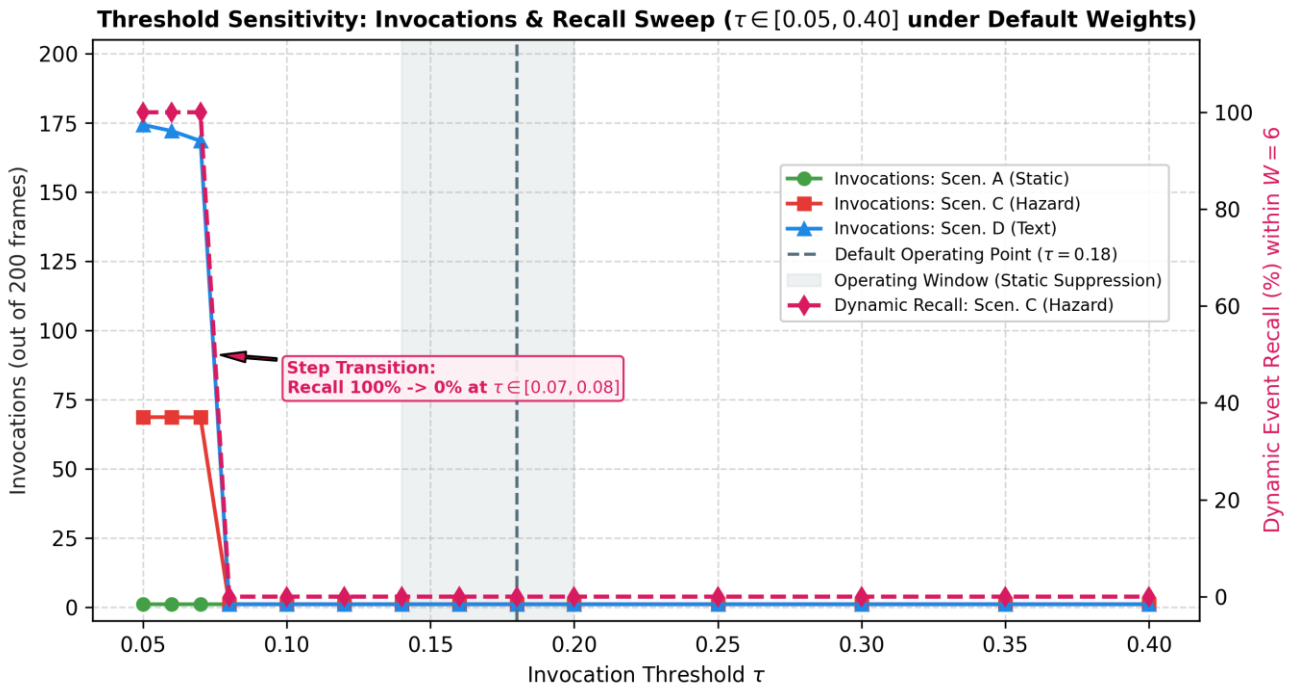


Fig. 5. Threshold sensitivity sweep (tau in [0.05, 0.40] under Default weights). The transition is a sharp step collapse between tau=0.07 and 0.08, rather than a gradual curve.

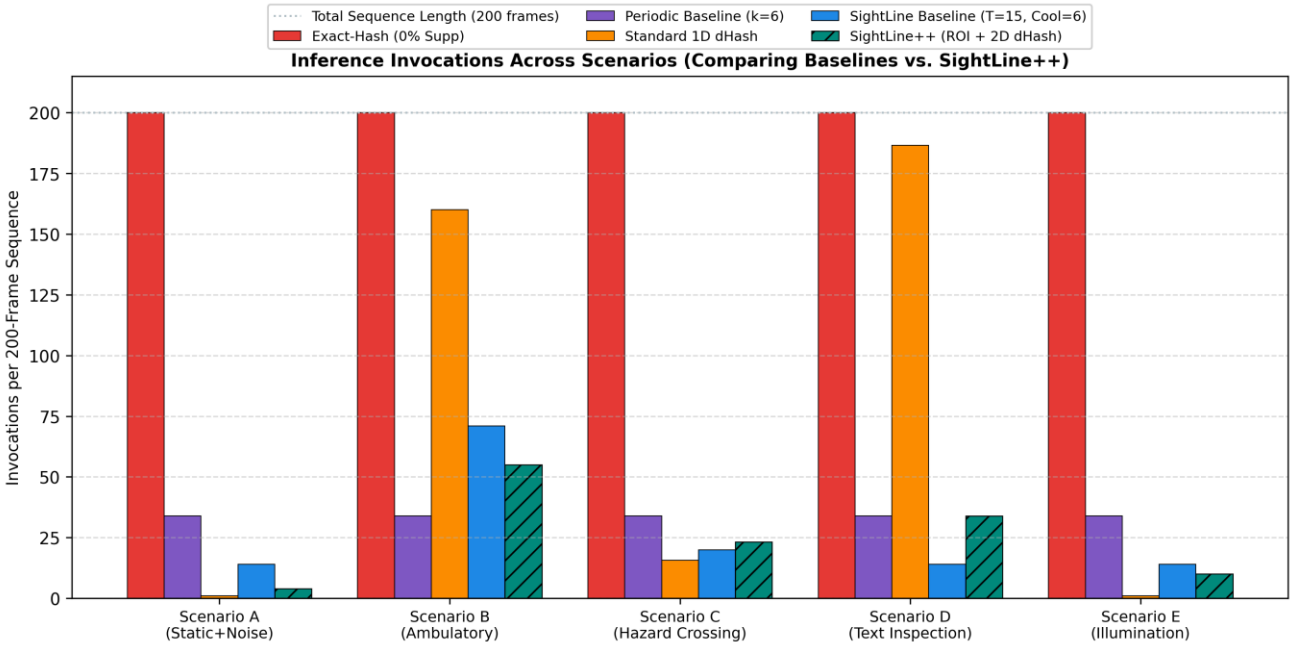


Fig. 6. Redundancy suppression and invocation counts across synthetic scenarios A–E (200-frame sequences). Exact perceptual hashing collapses under sensor noise on static views (0% suppression), while dead-zone filtering preserves suppression.

D. Exploratory Real-Video Evaluation

We processed one 20-s (600-frame) 1080p egocentric clip of workshop benchwork (head motion, cutters and grinders, analog sensor noise), downsampled to 30 fps. Florence-2-base FP16 was run on an RTX 2050 (4 GB; peak VRAM 618.9 MiB). Over 100 forward passes, <CAPTION> averaged 346.06 ms (322.5–367.7) and <OD> averaged 287.06 ms (275.0–312.9); the clip itself was processed with <CAPTION> and the hazard state inactive ($z_t = 0$), so the hazard pathway was not exercised on real video. The clip has no frame-level annotations, so real-world recall is unmeasured, and all budget-matched baselines were tuned on this same clip. Power and runtime come from Eq. (6)–(7) using Jetson Orin Nano parameters; the RTX 2050 supplies timing only.

TABLE III: REAL EGOCENTRIC CLIP (600 FRAMES, 20 S). POWER AND RUNTIME ARE MODELED.

Configuration	Invocations (supp.)	GPU duty	Modeled power	Runtime (32.56 Wh)
Continuous (30 fps)	600 (0.0%)	100.0%	11.50 W	2.83 h
Periodic k = 6	100 (83.3%)	100.0%*	11.50 W	2.83 h
Periodic k = 12	50 (91.7%)	100.0%*	11.50 W	2.83 h
Frame difference (tau = 0.22, budget-matched)	38 (93.7%)	69.4%	8.54 W	3.81 h
Exact 1D dHash (eps = 0)	600 (0.0%)	100.0%	11.50 W	2.83 h
1D dHash, tau = 0.18	138 (77.0%)	100.0%*	11.50 W	2.83 h
1D dHash, tau = 0.28 (budget-matched)	36 (94.0%)	66.0%	8.22 W	3.96 h
SightLine Endurance	32 (94.7%)	58.7%	7.52 W	4.33 h
SightLine++	117 (80.5%)	100.0%*	11.50 W	2.83 h

*GPU duty cycle reaches 100.0% under single-worker queuing because total service compute exceeds sequence duration (e.g., Periodic k=12: 50 x 413.41 ms = 20.67 s > 20.0 s).

Findings: (1) Exact hashing triggers on all 600 frames, confirming the need for the dead zone; (2) periodic sampling at 5 Hz (100 requests × 395 ms ≈ 39.5 s of work in a 20 s clip) and even 2.5 Hz (50 × 413 ms ≈ 20.7 s) saturate a single worker, so they save no energy under the duty-cycle model; (3) global motion stayed quiet (median $S_{\text{global_mad}} = 0.0084$, 95th percentile 0.0172, below the 0.02 active threshold), so the Endurance watchdog stayed at $T_{\text{ambient}} = 60$. Its 32 invocations comprise one startup call, approximately 9 heartbeats, and approximately 22 threshold-driven triggers; (4) budget-matched frame difference and 1D dHash land within 0.7–1.0 W of Endurance. The advantage over frame differencing is therefore mainly the reference-anchored drift accumulation, which this clip does not stress, rather than a large power gap. Single-clip results have no cross-seed variance and should not be generalized. Gating CPU time (2.03–3.45 ms) was measured on pre-downsampled thumbnails; software downsampling of 1080p frames added 138–156 ms, so deployment requires hardware ISP scaling.

VIII. COMPARISON, STRIDE SWEEP, AND SAFETY ANALYSIS

A. Periodic Sampling

For a periodic stride k and detection window $W = 6$, the phase-averaged recall is $min(k, W)/k$ and the mean delay for $k \geq 6$ is $(W - 1)/2 = 2.5$ frames. Recall falls from 100% ($k \leq 6$) to 75% ($k = 8$), 50% ($k = 12$), and 20% ($k = 30$). Under busy-skip scheduling with a 12-frame service time, $k = 6$ degrades to the $k = 12$ recall of 50%, because half of the requests are dropped.

B. Architectural Comparison with WorldScribe

WorldScribe [15] uses cloud VLMs for rich, query-directed narration; SightLine runs on-device and avoids network dependence, round-trip jitter, and cellular dead zones, at the cost of a smaller model and less lexical depth. WorldScribe is stronger for reading and conversational question answering. This is a comparison of architectures, not an empirical head-to-head, and we make no claim of superiority.

C. Stopping-Distance Budget

The nominal machine latency under an idle worker is 0.515 s (0 f gate + 380 ms + 135 ms). Adding human reaction and motor adjustment of 0.5–1.0 s gives a total of about 1.0–1.5 s, or 1.2–1.8 m of travel at 1.2 m/s. For a hazard closing at 4.2 m/s from 2.5 m (our speech example), the time to contact is about 0.6 s, which is shorter than the machine latency alone. We therefore do not claim that SightLine++ meets a safe-stopping budget. The system is a supplementary aid to a white cane or guide dog, not a collision-avoidance system. The latency figures also assume an idle worker: during sustained motion the active profile requests 5 Hz against ≈ 2.6 Hz capacity, so queueing delay would add to these numbers (not measured). The Endurance profile has a worst-case gate delay of 2.0 s (2.48 s total) and is intended for stationary or seated use only.

D. FMEA Risk Ledger

We assessed ten failure modes (localized hazard dilution, re-trigger storms, speech preemption delay, cooldown lockout, noise bit-flip collapse, sign invariance on documents, drift blindness, illumination bursts, low-contrast dormancy, quiescent hazard loss) using Severity, Occurrence, and Detection scores (1–10, SAE J1739 / AIAG-VDA style) that the authors assigned themselves. The summed Risk Priority Number falls from 2,904 to 264 after mitigations (mean RPN dropping from 290.4 down to 26.4). These ratings are subjective design estimates: they have not been validated by independent raters or field data, the reduction in some entries reflects a documented limitation (e.g., a disclaimer for document reading) rather than a technical fix, and the aggregate percentage should not be read as a measured reduction in safety risk. Residual risks include peripheral incursions under occlusion, masking of earcons in loud environments, thermal throttling above about 35°C, misclassification of walking as rest, and lens obstruction or very low light, where the VLM would receive an uninformative frame and the user would not know the camera is blinded.

IX. EDGE PLATFORM, ENERGY MODEL, AND PLANNED STUDY

A. Platform and Gating Cost

The target platform is the NVIDIA Jetson Orin Nano Developer Kit (6-core Cortex-A78AE, 1024-core Ampere GPU, 8 GB LPDDR5, 15 W mode) [16]. The Python/NumPy gate was timed on 160×120 thumbnails at 2.03--3.45 ms on a host Core i5-12500H, 2.45--3.50 ms on a Cortex-A78AE core benchmark, and 4.2--6.5 ms on a Cortex-A72-class core reference. That is under 11% of the 33.3 ms frame period on the first two platforms and about 13–20% on the third.

B. What Was Measured, Specified, and Modeled

Measured: gate CPU time; Florence-2 FP16 latency and VRAM on the RTX 2050; Piper TTS latency; Jetson idle power of 1.85 W (calibrated inline DC analyzer at the 19 V input, desktop and camera preview active, GPU idle).
Specified, not measured: active power of 11.50 W, taken from the vendor figure for the 15 W mode [16]; it was not measured with shunt resistors.
Modeled: average power and runtime:

$$P_{\text{avg}} = D_{\text{gpu}} P_{\text{active}} + \left(1 - D_{\text{gpu}}\right) P_{\text{idle}} \quad (6)$$

$$T_{\text{run}} = \frac{E_{\text{eff}}}{P_{\text{avg}}} \quad (7)$$

with $E_{\text{eff}} = 0.88 \times 37\ \text{Wh} = 32.56\ \text{Wh}$. The duty cycles D_{gpu} come from RTX 2050 service times. The Orin Nano has half as many GPU cores, so its true service times are probably longer, making our duty cycles and runtimes optimistic for this platform. No battery discharge logs were collected. All runtimes should be re-measured on the target hardware before being used as claims.

TABLE IV: SELECTED MODELED OPERATING REGIMES (ORIN NANO PARAMETERS, 37 WH PACK, 32.56 WH USABLE)

Regime	GPU duty	P_avg	Runtime (32.56 Wh)
Continuous / saturated configurations	100.0%	11.50 W	2.83 h
Mixed day, paced (30% active transit)	43.3%	6.03 W	5.40 h

Regime	GPU duty	P_avg	Runtime (32.56 Wh)
Mixed day, quiescent (70% standby, 30% active)	35.3%	5.26 W	6.19 h
Stationary ambient pacing (T_ambient = 60)	19.0%	3.68 W	8.84 h
Quiescent standby (5.0 s heartbeat)	7.6%	2.58 W	12.60 h

The mixed-day rows assume 70% standby and 30% active time, and the standby rows assume stationary use; these are scenario assumptions, not measurements. Under active locomotion the model saturates at 11.50 W (2.83 h).

C. Planned BLV Evaluation (Not Yet Performed)

We plan a within-subjects study with $N = 24$ BLV participants (12 legally blind, 12 low vision) under four conditions: no AI, continuous 30 fps streaming, periodic sampling ($k = 6$), and SightLine adaptive gating, ordered by a balanced Latin square, across an indoor mobility course, a 1.2 km urban route, and an indoor living/reading lab. Measures are collisions, walking pace, stopping latency, NASA-TLX [13], a 7-point masking rating, and the SUS [20]. With $N = 24$ and $\alpha = 0.05$ a paired t-test has about 80% power for $d = 0.60$; with six Holm-corrected comparisons per-test power drops to about 53%, and about 38 participants would be needed for 80%. The protocol is prepared for submission to an Institutional Review Board (IRB) and has not been approved or run; no human-subject results are reported here.

X. LIMITATIONS AND DISCUSSION

1. Limited real-world evidence: One 20-s clip of a static bench, no annotations, baselines tuned on the same clip, and no hazards. Outdoor pedestrian transit is a hypothesis.
2. Synthetic benchmarks are scripted: Recall and delay values are consistency checks under tuned thresholds. The cleanest claim is the qualitative one: global gating misses localized hazards, and ROI pooling restores sensitivity.
3. Active navigation reduces to periodic sampling: During motion the watchdog dominates and the 5 Hz request rate exceeds worker capacity, so energy savings come from stationary or quiet periods. Hybrid use with an IMU for mode switching is future work.
4. Untested hazard pathway: The detector-driven hazard state, the person rule, and the speech queue were not evaluated on real data or users.
5. Modeled energy projections: Energy numbers are modeled from a different GPU than the target and use a vendor active-power figure.
6. Document reading limitations: Documents need OCR or a validated two-sided edge filter; hash-based gating alone performs at chance level there.
Future work: An onset-phase sweep isolating ROI pooling from the watchdog; evaluation on Ego4D [17], EPIC-Kitchens [18], and VizWiz [19]; measurement on the physical Jetson; and the user study once approved.

XI. CONCLUSION

SightLine gates a wearable VLM by comparing frames with the last processed reference, which accumulates drift and, with a dead-zone hash, avoids the noise-driven collapse of exact hashing (0% suppression on real video). Its two main profiles sit on opposite ends of a safety–efficiency trade-off: Endurance suppresses 94.7% of frames on a quiet workbench, with a modeled 4.33 h runtime, but is blind to localized hazards, while SightLine++ detects localized incursions in simulation but saturates a single worker during continuous manipulation. Strong claims about safety and battery life await measurement on the target hardware, hazard-bearing real video, and the planned user study.

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APPENDIX: GATE PSEUDOCODE

```
Algorithm 1: Reference-Anchored Multi-Cue Scene-Change Gating (SightLine)
Input: Incoming video stream I_t at 30 fps; task profile (alpha, beta, gamma, delta, tau, T_ambient, T_active,
T_cool)
Output: Binary gating decision D_t in {0, 1}; updated reference frame I_tp
State: Reference frame I_tp, hazard latch z_t in {0, 1}, last alert time t_alert, last invocation time t_p

1.  if t == 0: return D_t = 1, I_tp = I_0  // Startup trigger
2.  Compute dead-zone dHash: h(I_t) = [B_H || B_V] (2D) or B_H (1D) with eps = 7.0 DN
3.  Compute visual change: S_vis = HammingDistance(h(I_t), h(I_tp))
4.  Compute appearance change: S_app = 1.0 - CosineSimilarity(Hist_RGB(I_t), Hist_RGB(I_tp))
5.  Compute 4x4 spatial ROI motion: S_roi_max = max_{r,c} MAD(Block_{r,c}(I_t, I_tp))
6.  Compute motion: S_mot = (MAD_global(I_t, I_tp) if Endurance else 0.50 * MAD_global + 0.50 * S_roi_max)
7.  Update watchdog: T_max = (T_active if z_t == 1 or S_mot >= 0.02 or S_roi_max >= 0.05 else T_ambient)
8.  if (t - t_p) >= T_max: return D_t = 1, I_tp = I_t  // Watchdog timeout
9.  if S_roi_max >= 0.35 or S_mot >= 0.20: return D_t = 1, I_tp = I_t  // Motion override (Eq. 3)
10. Downstream VLM update (when D_t == 1): z_t = 1 if any(IsHazard(o_i) for o_i in Entities(I_t)) else 0
11. S_haz = (z_t if (t - t_alert) >= T_cool else 0.0)
12. S_t = alpha*S_vis + beta*S_app + gamma*S_mot + delta*S_haz
13. if S_t >= tau and (t - t_alert) >= T_cool: return D_t = 1, I_tp = I_t  // Nominal trigger
14. return D_t = 0  // Frame suppressed (<3.5 ms CPU compute)
```